ROBERT BAIN

COMPUTER VISION | MACHINE LEARNING | SEMICONDUCTOR ENGINEERING

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ENGINEERING PROFILE

Computer vision and software engineer with an M.S. in Artificial Intelligence and hands-on microfabrication expertise. Built real-time perception for autonomous systems, optimized inference for embedded hardware, and engineered a patent-pending neural recording platform from wafer fabrication through electronics and analysis.

TECHNICAL STRENGTHS

Computer vision: Object detection, image segmentation, image registration, camera calibration, stereo depth, 3D tracking, pose estimation, attention mechanisms, geometric vision

ML and software: Python, C++, PyTorch, TensorFlow, OpenCV, MATLAB, Java, JavaScript, SQL, Linux, Git, ROS, transfer learning, data augmentation

Deployment and optimization: ONNX, TensorRT, CUDA, Jetson AGX Xavier, Coral Edge TPU, model pruning, activation sparsity, embedded inference

Semiconductor and hardware: Photolithography, photomask design, thin-film polymers and metals, flexible circuit design, PCB design, KiCad, Fusion 360, AutoCAD

ENGINEERING AND RESEARCH EXPERIENCE

Graduate Researcher | UC Irvine **(2023 to 2026)**

- Invented patent-pending flexible neural probes using lithography and thin-film polyimide and metals. Developed scalable wafer fabrication and simplified assembly.
- Reduced estimated materials and assembly costs to ~$12 per probe versus ~$500 reported for an earlier Nature Communications design. Yielded >10× more recording channel density per unit body mass.
- Designed compact acquisition electronics, PCBs, connectors, and packaging. Integrated fabrication, circuits, data acquisition, and software into a working research platform.
- Built GPU-accelerated Python data pipelines and validated three-month recordings. Standardized manufacturing and mentored five undergraduates in assembly, QC, and troubleshooting.
- Co-developed pairwise learning to improve image regression and classification from limited labels. Evaluated transfer learning and data augmentation. First-author paper submitted to Machine Learning with Applications.
- Reported state-of-the-art facial age prediction with 3.64-year average error, using a compact model trained from scratch on one consumer GPU.
- Developed interpretable and compute-efficient 3D MRI vision models to predict cognitive traits with limited data and identify informative brain regions.
- Built multi-camera acquisition, calibration, and depth-based 3D pose tracking tools using OpenCV and DeepLabCut for behavioral experiments.

Computer Vision Engineer, Autonomous Racing | Global Formula Racing, Oregon State **(2019 to 2021)**

- Built real-time cone detection and geometric perception for an autonomous racing car, using homographies and Perspective-n-Point. Developed camera-LiDAR co-registration to align visual and depth measurements.
- Optimized and deployed vision models with ONNX and TensorRT on NVIDIA Jetson AGX Xavier for an international autonomous racing competition.

ADDITIONAL SOFTWARE AND AI EXPERIENCE

Robotics Researcher | Cognitive Anteater Robotics Lab, UC Irvine (2022 to 2023)

- Developed camera-based localization and ROS multisensor navigation for Toyota HSR and Clearpath Jackal, including camera, LiDAR, IMU and odometry transforms. Used ONNX and TensorRT for HSR inference.
- Worked on CARLsim6, a C++ and CUDA spiking neural network simulator, and developed Git-based continuous integration workflows.
- Implemented camera-image localization with PyTorch, including custom datasets, training, and position estimation for indoor robots.

Machine Learning Research, Model Compression | Oregon State University (2020 to 2022)

- Used iterative magnitude pruning to train models with 33x fewer active weights while preserving accuracy. Studied robustness to noisy inputs, activation sparsity, and continual learning.

AI Researcher, DARPA Machine Common Sense | Oregon State (2020 to 2021)

- Sole developer of the Theory-of-Mind component and led a separate team effort for DARPA Machine Common Sense. OSU placed first in two of three evaluation categories.

Lead Software Developer | Northwest Media, Eugene, Oregon (2018 to 2020)

- Led development of web software serving approximately 150,000 users and public-sector clients. Supervised two junior developers and modernized 68 interactive training courses from Flash to HTML5.
- Developed custom reporting software for government clients and maintained production training services supporting large public-sector programs.

Full-Stack Software Developer | Contract Projects, Eugene, Oregon (2018)

- Built and deployed public websites for regional organizations, including Cascadia Prepared and Alignment Ventures.

SELECTED EMBEDDED AI PROJECTS

- Built a Surface Pro 4 system that autonomously played Guitar Hero III at 60 FPS, with CNN inference on a Google Coral Edge TPU, screen capture, controls, and OBS recording. Demo

EDUCATION

University of California, Irvine: M.S., Neurobiology and Behavior, September 2026. GPA: 4.0

Oregon State University: M.S., Artificial Intelligence, 2022. GPA: 3.6

Oregon State University: B.S., Biology, Chemistry minor, 2016

SELECTED TECHNICAL WORK

Multi-camera imaging | Visual localization | GPU data processing | Pairwise learning code

PATENT AND RESEARCH PUBLICATIONS

U.S. PROVISIONAL PATENT

U.S. Provisional Patent Application No. 64/130,528 (2026). *Capillary-force-assembling, Stiffener-free, High Channel Density Neural Probes.* Filed August 10, 2026. Patent pending.

PEER-REVIEWED JOURNAL PAPERS

Shobe, J. L., Ghanbarian, E., Bain, R., Saxena, R., Chandrasekaran, M., & McNaughton, B. L. (2025). *Cognitive Enrichment Improves Spatial Memory and Alters Hippocampal Synaptic Connectivity in a Mouse Model for Early-Life Stress.* Frontiers in Cellular Neuroscience, 19, 1646883. DOI

Espino, H., Bain, R., & Krichmar, J. L. (2024). *A Rapid Adapting and Continual Learning Spiking Neural Network Path Planning Algorithm for Mobile Robots.* IEEE Robotics and Automation Letters, 9(11), 9542-9549. DOI

Espino, H., Bain, R., & Krichmar, J. L. (2023). *Selective Memory Replay Improves Exploration in a Spiking Wavefront Planner.* International Joint Conference on Neural Networks (IJCNN). DOI

2026 MANUSCRIPTS

Bain, R., Rashidi, M., & Chrastil, E. R. (2026). *Pairwise Learning Increases Data Efficiency in Deep Neural Networks.* Submitted to Machine Learning with Applications, October 2026. Manuscript PDF

Rashidi, M., Bain, R., et al. (2026). *Inferring Continuous Cognitive Traits from Limited Neuroimaging Data Using Pairwise Convolutional Neural Networks.* Manuscript, 2026. Manuscript PDF

PREPRINTS

Bain, R. (2022). *The Role of Biology in Deep Learning.* M.S. thesis research, arXiv:2209.04425. arXiv

Lally, A., Bain, R., & Alotaibi, M. (2021). *ePose: Let's Make EfficientPose More Generally Applicable.* arXiv:2111.15114. arXiv

Bain, R. (2021). *Visualizing the Loss Landscape of Winning Lottery Tickets.* arXiv:2112.08538. arXiv